

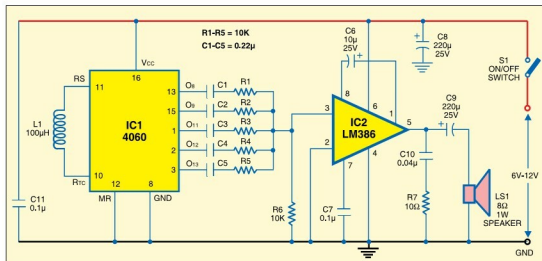
MULTITONE SIREN

■ PRADEEP G.

This multitone siren is useful for burglar alarms, reverse horns, etc. It produces five different audio tones and is much more ear-catching than a single-tone siren.

The circuit is built around popular CMOS oscillator-cum-divider IC 4060

and small audio amplifier LM386. IC 4060 is used as the multitone generator. A 100µH inductor is used at the input of IC 4060. So it oscillates within the range of about 5MHz RF. IC 4060 itself divides RF signals into AF and ultrasonic ranges. Audio signals of different frequencies are available at pins 1, 2, 3, 13 and 15 of IC 4060 (IC1).



These multifrequency signals are mixed and fed to the audio amplifier built around IC LM386.

The output of IC2 is fed to the speaker through capacitor C9. If you want louder sound, use power amplifier TBA810 or TDA1010.

Only five outputs of IC1 are used here as the other five outputs (pins 4 through 7 and 14) produce ultrasonic signals, which are not audible.

Assemble the circuit on a general-purpose PCB and enclose in a suitable cabinet. Regulated 6V-12V (or a battery) can be used to power the circuit. ●

PHOTOMETER

■ D. MOHAN KUMAR

An illuminance of 100 to 1000 lux is required for reading and doing close work without eyestrain. Specular illumination or bright sunlight provides 50,000 lux, while twilight or dim light provides only 10 lux. Reading and close work under a fluorescent lamp is better because it can give a flux of 4400 lumens in contrast to 1600 lumens of a tungsten lamp. If the eyes are exposed to dim light for many hours, dark adaptation will develop and there will be severe eyestrain.

Here is a circuit for measurement of light intensity. It can give a visible linear scale of illuminance in a

comparative form of different light levels such as minimum, moderate and maximum.

The circuit uses a p-n photodiode as the light sensor and a display driver IC LM3915 (IC1) to give a logarithmic current scale linearly with the incident light energy. The photodiode has a high frequency response and fast switching speed of one nanosecond compared to cadmium-sulphide (CdS) photodetectors. The 'n' material of the photodiode is composed of pure silicon doped with phosphor to give free electrons. The 'p' material is formed of pure silicon with boron as the doping material. It provides free holes.

In dark, reverse saturation current develops in the photodiode and free

electrons move when it gets light. The linearity of light current generation in the photodiode is proportional to irradiance. Typically, the light current is 10 to 90 µA at 5 volts and the radiation sensitivity of a p-n photodiode is 0.4 to 4 µA/mW/cm².

In the circuit, the photodiode is connected in forward-biased mode through current-limiting resistor R1 to get the light current. The current generated by the photodiode in response to light passes to input pin 5 of IC1 through preset VR1.

IC LM3915 has a complete built-in voltmeter circuit to indicate varying input voltage directly on

